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The impacts of carbon pricing on the electricity market in Japan

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To achieve carbon neutrality by 2050, Japan should speed up reducing fossil fuel reliance on production, especially for energy-intensive sectors. One way is by implementing a carbon pricing system, converting emissions from fossil fuels to costs of production and consumption. This study focuses on the correlation between the price of wholesale electricity spot market and carbon cost of nine regions in Japan through carbon cost pass-through rate. This paper applies polynomial OLS regression with degree two through machine learning technics to better fit the relationship between electricity price and demand and also applies a generalized additive model to capture the nonlinear relationship between fuel spread and carbon cost and test the robustness of estimated CPTR. The results show that Hokuriku, Kyushu, Shikoku, Tohoku, and Tokyo have a lower value carbon cost pass-through rate while Kansai, Chubu, and Chugoku have a higher rate of carbon cost pass-through. There is a special case in Hokkaido as the negative relationship between electricity price and carbon cost. Those findings are also crucial in supporting future policy adjustments.

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Introduction

The Paris Agreement in 2015 sets out a long-term goal to keep the global temperature increase below 2 °C above the pre-industrial levels and pursue efforts to keep it to 1.5 °C. One way to control temperature is to reduce greenhouse gas emitted into the atmosphere. Carbon dioxide (CO_2) comprises 74% of greenhouse gas emissions. Most CO_2 emissions (93%) are from fossil fuels, and thus it is a long-term transition for countries and regions from high-carbon to low-carbon economies. Electricity generation accounts for the highest carbon emissions among energy usage sectors. Therefore, the carbon neutrality plan's emissions reduction in the electricity sector is crucial. As the third-largest GHG emitter globally, Japan has committed to achieving carbon neutrality by 2050. To reduce CO_2 emissions, the Ministry of the Environment implemented the experimental Japan Voluntary Emissions Trading Scheme (JVETS) in 2005; the rule of voluntary participants led to only a minor impact on CO_2 emissions. Instead of establishing a national cap-and-trade carbon market, Japan introduced a carbon tax in 2002. The high level of CO_2 now mainly results from the low-carbon tax rate. This research analyzes the impact of the carbon pricing system on electricity prices at the regional level in Japan.

According to IEA 2019, the electricity and heating sector accounts for about 48% of CO_2 generated, and around 70% of electricity and heating in Japan is from thermal plants. 20% is from renewable sources, and the rest are from nuclear (6%) and other sources (4%). The dependency on nuclear dropped dramatically after Japan's nuclear disaster in 2011. Therefore, carbon emissions in electricity generation are derived mainly from fossil fuels burning for thermal plants. In Japan, electric utilities comprised of 1 Generation and Transmissions company and 9 Distribution companies since 1939. After 1951, companies in each area (Hokkaido, Tohoku, Tokyo, Hokuriku, Chubu, Kansai, Shikoku, Chugoku, Kyushu) reformed to vertically integrated utilities, which refers to General Electric Utilities, GEUs. Japan Electric Power Exchange, JEPX is an exchange to facilitate electrical power transactions among power utilities. It commenced trading in wholesale power transactions after April 1, 2005. Along with the carbon pricing system, the retail electricity market in Japan was opened up for competition in April 2016 (Chapman, 2018).

Theoretical backgrounds and methods

Theoretical backgrounds. One way to solve the negative externalities problem is to internalize air pollution by adequately pricing emissions (Pigou, 1920). The purpose of carbon pricing was to encourage producers to adopt renewable sources by increasing the costs of fossil fuel combustion (Nazifi, 2016). A well-design carbon pricing mechanism plays a role in efficiently passing the cost of CO_2 emissions on generators (Liu, 2012). Effective carbon pricing would pressure on producers directly, increasing marginal cost as emissions intensity arises from fossil fuel combustion. Producers can transfer a certain proportion of the burden to consumers by raising the price of products.

Based on the findings of Bauer and Zink (2005), the power price is determined by the trend in CO_2 price only. Sijm and Chen (2006) and Chen (2008) assumed that electricity was determined by the variation of fuel and carbon costs, and thus they introduced a linear regression model for analyzing the impact of the carbon price on electricity prices. Some research finds that the liberalization of electricity is a prerequisite for carbon pricing systems to avoid a shortage of electricity supply (Fan, 2014). Under the economic theory of a competitive market, the marginal revenue should be equal to the marginal cost. Therefore, if other costs are constant, the difference between marginal revenue and marginal fuel cost would be mainly explained by the cost of carbon allowance for electricity generation multiples the carbon pass-through rate. However, it ignores the relationship between demand/supply and prices. Further studies (Barlow, 2002) applied a structural model (Kanamura, 2007) to find out if there is a nonlinear relationship between electricity price and electricity demand. Therefore, in this paper, electricity price in a competitive market was influenced primarily by electricity demand, carbon intensity and pricing mechanism[1]. Electricity price rises as carbon cost increases, but the degree of price raised demonstrates the possibility that firms reduce fossil fuel reliance in the process of production. The carbon pricing system plays a role in reducing CO_2 emissions by the rising production cost of firms based on their carbon intensity. However, the effectiveness of the carbon pricing policy is impaired by the fact that firms have abilities to switch the burden of cost onto consumers, especially for inelastic products. Given other factors constant, if the extent of electricity price increased more than the carbon cost increased, generators would not have enough pressure to transform from high-carbon to low-carbon production, and the welfare of consumers would reduce as the price increased. This paper focuses on two points on the results of empirical analysis in Japan. The first is the correlation between electricity price and carbon cost. Typically, there is a positive relationship between electricity price and carbon cost. In other words, the carbon cost increases along with the electricity price increases. The second is the value measures the degree of electricity price variations attributed to carbon cost. From the empirical analysis in Australia (Nazifi, 2016) and Spanish (Fabra, 2014), a carbon tax is typically borne more by customers than producers, which means the rates in the empirical analysis are usually higher than the situation when generators took more tax cost burden.

Indicator. The carbon cost pass-through rate (CPTR) is a commonly used indicator to quantify the degree of carbon emissions cost transferred to consumers. CPTR represents the ratio of changes in electricity prices to the changes in the marginal cost of carbon pricing. (Sijm, 2012). Estimating CPTR could help to reflect the extent of carbon cost in electricity production passing to consumers through a higher electricity price. This paper refers to Nazifi et al. (2021) methodology by applying the carbon costs pass-through rate (CPTR) to measure the extent to of electricity price is ascribed from carbon cost variations. Based on the assumption of a competitive market structure and at the equilibrium point of each electricity price, the carbon pass-through rate should be around 100%. In other words, the marginal fuel spread for wholesale electricity generators is supposed to variate proportionalities to the carbon cost introduced by the carbon pricing mechanism. However, many research results indicate that the carbon pass-through rate usually deviates from 100%. Nazifi et al. (2021), in their paper, show that three out of four regions in Australia have over a unit CPTR. Simshauser (2007) conclude that Australia's CPTR is almost 393%. Natalia et al. (2014) find that the emission costs are almost passed through electricity prices in Spain, and the reason is that electricity demand in Spain is inelastic demand, and meanwhile under the high-frequency auctions for electricity.

The fuel data period in this paper ranges from April 2016 to August 2021, when the carbon tax in Japan has no further adjustments. There is a limited research focus on CPTR based the empirical analysis at the regional level in Japan. The contribution of this research is to apply hourly data to analyze the effect of carbon pricing on electricity spot market prices. The carbon cost

pass-through rate is defined as the following formula:

$$\rho = \frac{dP/P}{dC/C} \quad (1)$$

where P is electricity price, and C is carbon cost. A higher rate demonstrates that customers in the spot market take more burdens on carbon emissions costs than generators as one unit increase of cost leads to an increase in one more unit of electricity price. The higher rate of CPTR demonstrates that the increase in marginal revenue for generators largely offsets the cost increase induced by the carbon pricing mechanism. A higher carbon pass-through rate means that the consumers carry higher proportions of carbon costs. Although this paper only focuses on the carbon pass-through rate in the wholesale electricity price, and the retail price paid by the consumers would be higher, a higher carbon pass-through rate in wholesale price can also indicate that retailers have a greater potential to transfer the higher price to consumers than the case of a lower CPTR. On the contrary, a lower rate means generators bearing high pressure on carbon cost, which may stimulate producers to reduce costs by using energy efficiently or raising investments in low-carbon assets such as renewable energy facilities. Therefore, electricity generators in areas with a higher carbon pass-through rate have a larger possibility of producing renewable resources. Analysis of carbon cost pass-through rate in each region of Japan supports policy-makers in adjusting decarbonization strategies.

Regression results and discussions

Data and feature analysis. In 2011, the Great East Earthquake and the Fukushima Nuclear Power Plant accident induced a shoot-up in electricity prices, especially in Tokyo and Tohoku. These accidents strengthen the pressures for electricity reformation. There are two objects under the reformation. The first is to reduce the electric price, and the second is to establish a stable power supply, maintaining the demand-supply balance. Electricity prices decreased and were stable below 30 JPY per kWh in all regions since the electric system reformation in 2013, and the fluctuation of price also reduced.

Figures 1 and 2 together illustrate electric spot market prices of nine regions in Japan: Hokkaido, Tohoku, Tokyo, Hokuriku, Chubu, Kansai, Shikoku, Chugoku, and Kyushu from the period this paper focused on, which is from January 2016 to August 2021. The electricity prices in Japan hit a high record in January 2021 since the cold grips the country and demands for heating increased abruptly. Figure 2 shows that the electricity price is above 140 JPY per kWh at the highest point. The prices recovered under 20 JPY/kWh in a month after the electricity supply returned to normal.

Combing the results of described statistics in Table 1, among those areas, the fluctuation in Hokkaido is tremendous, and the average electricity price is 5.14 JPY per kWh, which is also the highest. The average price in Tokyo is about 3.144 JPY per kWh, and in Tohoku is 3.95 JPY per kWh. The variations mode in those two areas is approximate, but Tokyo reaches a higher price at extreme times. The average electricity price in the rest regions is below 2 JPY per kWh. The average price in Kansai is 0.737 JPY per kWh, which is the least.

If the electricity price can reveal a carbon tax, it would increase as carbon costs arise. From Table 2, carbon intensity in Hokkaido is also highest and way higher than in other areas, which is 0.43 kg-CO_2/kWh. The standard deviation of Hokkaido is the largest, and thus fluctuations of carbon intensity are tremendous. For the rest regions, the positive relationship between electricity price and carbon cost is not straightforward from a mean value.

Figure 3 depicts a scatterplot and smooth line of electricity spot market price and hourly electricity demand, indicating a nonlinear relationship between the two variables.

Therefore, to better analyze the variation of electricity price, this paper introduces a polynomial relationship of degree two between electricity price and demand by k-fold cross-validation to determine the optimal fitting degree and assumes that the wholesale electricity can be attributed to the variation of cost of fuels, demand for electricity and cost of carbon emissions. The electric demand captured the seasonality variation of the electricity spot price. Fuel costs dominate the marginal cost of generation of electricity. Labor and maintenance costs are negligible, or zero compared with fuel costs. Therefore, the other electricity generation costs, such as labor and maintenance costs, are assumed to be constants in the model. Other factors, such as technology, and market structures, should also be constant. The assumption is tenable in this paper as those factors would not variate much in the short term. It also assumes that fuel costs are fully and directly passed on to electricity prices.

The carbon pass-through rate estimated by the polynomial ordinary least squares (OLS) regression is defined as the following formula:

$$P_i^e - P_i^f = \alpha_0 + \alpha_1 C_i + \beta_1 D_i + \beta_2 D_i^2 + \varepsilon \quad (2)$$

The left-hand side of the equation is the spark and dark spread[2], which is the difference between the power price (JPY/kWh) and the cost of fuel (JPY/kWh) needed to produce that electricity. P_i^e is their hourly electricity spot market price from JEPX, respectively (i: 1,2,3…9) as the price on day-ahead spot market depends on single price auctions, which determined by trading volume. P_i^f refers to the marginal input costs, which are the costs of fuel needed to produce an additional unit of electricity. The fuel cost refers to the short-run marginal cost measures the cost to produce a unit of electric energy (not power), given an existing power plant. The marginal fuel cost of a plant that uses coal, oil, or natural gas is determined by the plant's efficiency or "heat rate," which is the ratio of input energy to output energy. It is calculated by the heat rate (MMBtu/kWh) multiple of the market price of fuel (JPY/MMBtu) as a portion of the short-run marginal cost. Fuels prices included the price of gasoline, light oil, kerosene, and coal. The first three categories of data are from Japan Agency for Natural Resources and Energy. The coal consumption in Japan heavily relies on imports and based on reports of MEI, the proportion of coal imports is almost 99% in 2019. Therefore, the import price of coal represents the price of coal. The final calculations of fuels price considered the combustions proportion of each fossil fuel for thermal plants. D_i refers to the hourly electricity demand amount in each region. C_i stands for the carbon emissions cost. The most challenging part of calculating the carbon cost pass-through rate is to quantify the carbon cost, which is non-observable. In this paper, the carbon emissions cost depends on the carbon tax rate multiple of the carbon intensity. Japan introduced a carbon tax in 2002 for carbon pricing, and the final tax rate equals JPY 289 per ton of CO_2 after several adjustments, and there has been no further increase since April 2016. Therefore, the carbon price in this paper is constant, and it is not hour-based data. The carbon emissions intensity equals total carbon emissions in thermal plants divided by total electricity generations. Besides, the main target of this paper is to analyze and compare the carbon pass-through rate in each region of Japan, and the electricity market is localized. It cannot be transported at very speeds. The electricity price and demand have a robust regional specification. Therefore, the trading in each region would not have many influences on prices in another region. The final carbon emissions cost is hourly

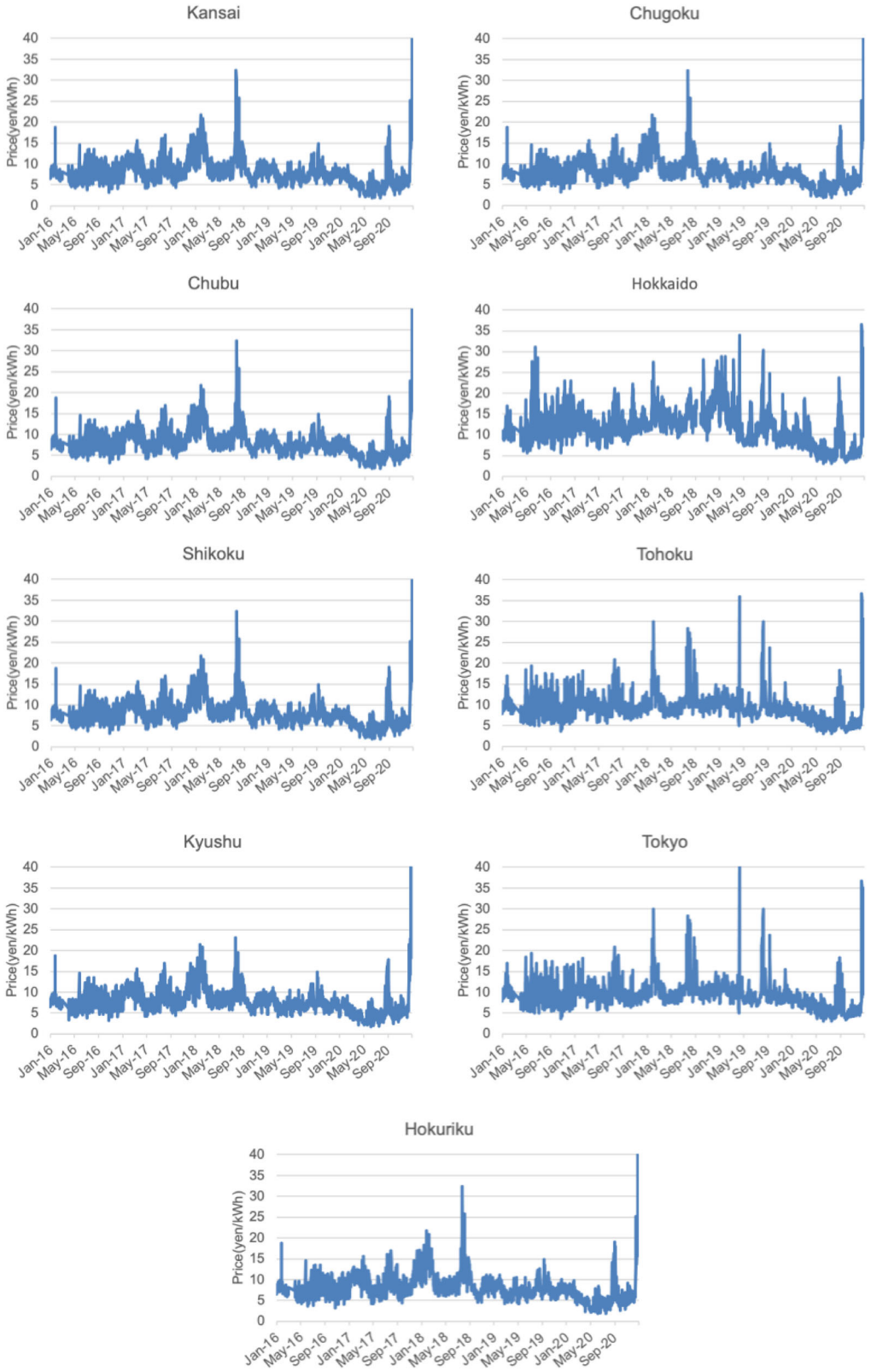

Fig. 1 Daily volume-weighted spot electricity price from 1 January 2016 to 31 December 2020. From the top left to the bottom right are variations of the electricity price for regions of Kansai, Chugoku, Chubu, Hokkaido, Shikoku, Tohoku, Kyushu, Tokyo, and Hokuriku.

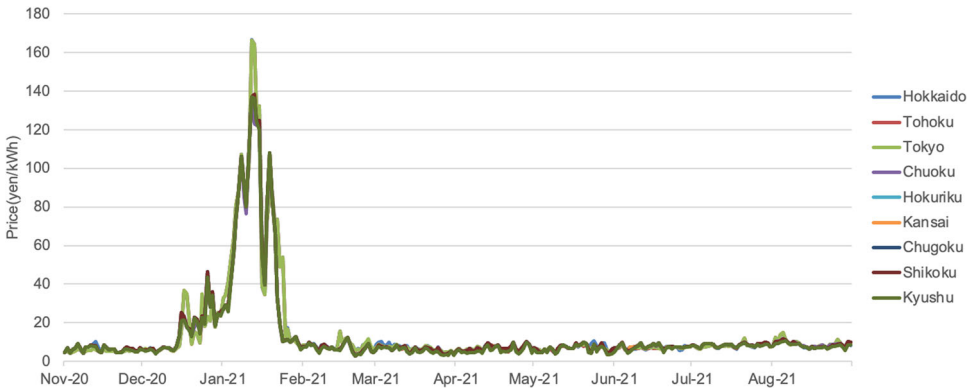

Fig. 2 Electricity price proliferated at the beginning of 2021. The electricity spot market price hit a high record in Japan in January 2021 since the cold grips the country.

Table 1 Summary statistics of electricity price in each region.

Regions	N	Mean	Std. dev.	1st Perc.	99th Perc.	t-value
Kansai	47,496	1.996	10.503	−4.913	37.387	41.409
Chugoku	47,496	0.737	10.504	−6.299	36.173	15.29
Chubu	47,496	1.967	10.411	−4.879	34.435	41.181
Hokkaido	47,496	5.140	11.712	−2.826	41.007	95.643
Shikoku	47,496	1.961	10.528	−5.349	37.34	40.595
Tohoku	47,496	3.095	11.346	−2.808	42.291	59.457
Kyushu	47,496	1.596	10.495	−7.318	35.344	33.139
Tokyo	47,496	3.144	11.357	−2.571	42.526	60.327
Hokuriku	47,496	1.964	10.502	−5.032	37.342	40.747

Table 2 Summary statistics of carbon intensity in each region.

Regions	N	Mean	Std. dev.	1st Perc.	99th Perc.	t-value
Kansai	47,496	0.15	0.02	0.08	0.22	1130.42
Chugoku	47,496	0.17	0.02	0.10	0.23	1457.13
Chubu	47,496	0.13	0.02	0.06	0.16	1182.41
Hokkaido	47,486	0.43	0.20	0.13	0.79	394.71
Shikoku	47,496	0.17	0.03	0.09	0.25	1015.64
Tohoku	47,496	0.16	0.02	0.10	0.22	1485.7
Kyushu	47,496	0.14	0.03	0.06	0.23	824.48
Tokyo	47,496	0.13	0.01	0.09	0.16	1765.06
Hokuriku	47,496	0.16	0.03	0.08	0.24	960.57

data as the carbon intensity variates in an hour. The price of electricity, fossil fuels and carbon are all converted to the unit of JPY per kWh to facilitate further analysis.

$$\alpha_1 = \frac{dlnP}{dlnC} = \frac{dP/P}{dC/C} = \rho \quad (3)$$

The α_1 is the carbon cost pass-through rate being focused on. This paper applies logarithm transformation for variables. The values of α_1 in the polynomial OLS regression function are equal to ρ.

From Table 3, all the results are statistically significant at a 1% level, which means that each region's carbon pass-through rate on electricity spot price is significant. The magnitude indicates the proportions of the consumers and producers bearing the carbon tax. In Kansai, Chubu, and Chugoku, one unit of carbon cost increase resulted in 1.934, 2.157, and 2.995 units of increased electricity price, respectively, meaning consumers in those regions paid more and undertook a higher rate of carbon cost than electric generators. Consumers in Chugoku bear the most proportions of the carbon tax. On the contrary, in Hokuriku, Kyushu, Shikoku, Tohoku, and Tokyo, the carbon through rate is 0.446, 0.531, 0.226, 0.585, and 0.698, respectively. Producers in those areas take a higher burden on carbon emissions cost, increasing the possibility of reducing carbon emissions to maintain net profits. Electric producers in Shikoku bear the highest proportion of carbon emissions cost. However, there is one exception among those returns. Hokkaido's negative carbon cost pass-through rate stands for a negative relationship between carbon cost and electricity price, indicating the increment of carbon cost may come along with electricity price reduction. Japan introduced a Feed-in tariff system in 2012 to encourage renewable resource use, and Hokkaido is a suitable area for wind power generation. One way to interpret the counterintuitive value is that the feed-in tariff from the local government offset the electricity price increase resulting from the fuel price rise. Therefore, Hokkaido's spark and dark spread have a higher proportion of negative values than other areas in the database.

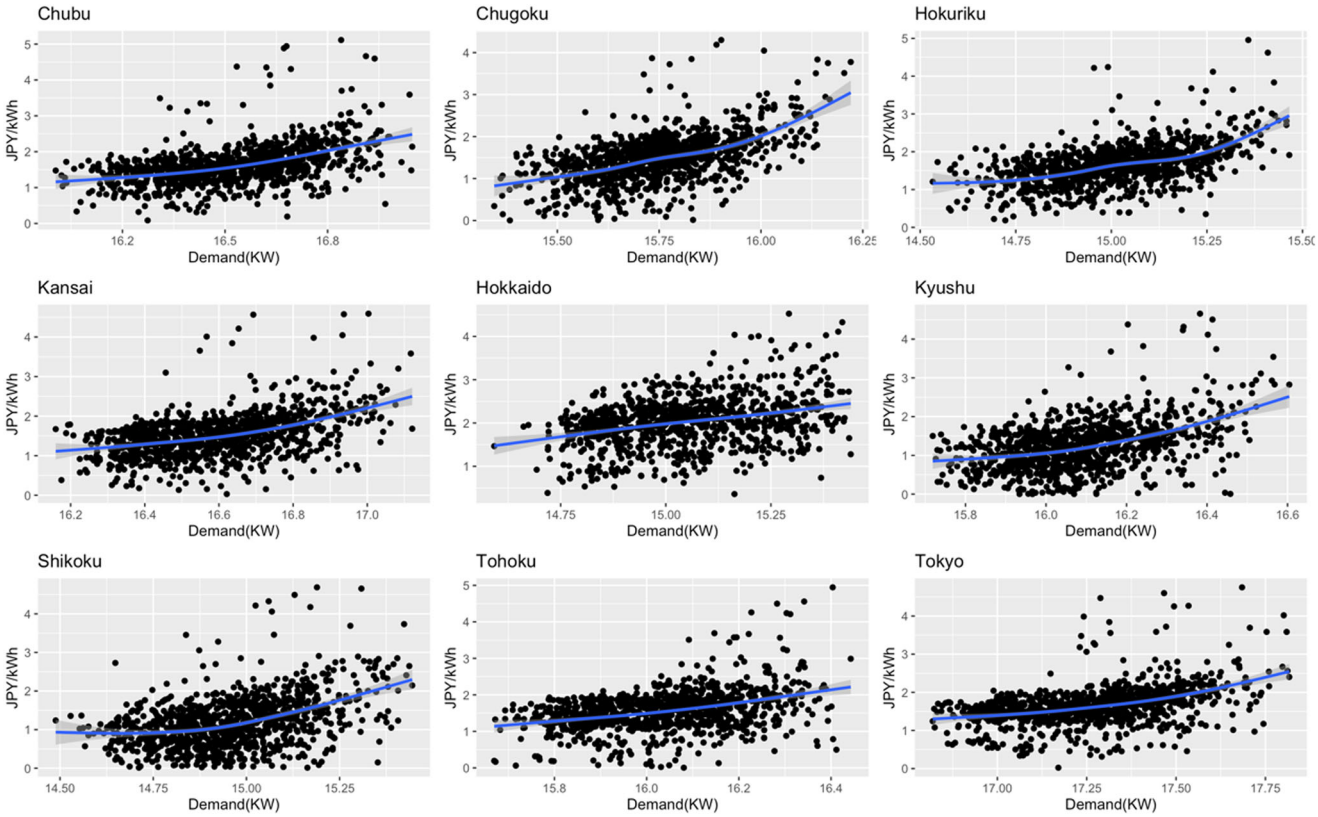

Fig. 3 Relationship between hourly electricity spot market price and hourly electricity demand in Chubu, Chugoku, Hokuriku, Kansai, Hokkaido, Kyushu, Shikoku, Tohoku, and Tokyo. The scatterplot and smooth line capture the non-linear relationships between electricity price and demand in nine regions.

Table 3 Regression results for the polynomial regression model $P_i^e - P_i^f = \alpha_0 + \alpha_1 C_i + \beta_1 D_i + \beta_2 D_i^2 + \varepsilon$.

Regions	α_1	β_1	β_2	α_0	R^2
Kansai	1.937*** (0.038)	23.721*** (6.037)	−0.581*** (0.181)	−229.462*** (50.384)	0.365
Chubu	2.157*** (0.053)	−62.266*** (5.576)	2.007*** (0.168)	485.687*** (46.264)	0.318
Chugoku	2.995*** (0.084)	135.768*** (13.353)	4.459*** (0.421)	1037.622*** (105.960)	0.291
Hokkaido	−0.055*** (0.003)	1.441*** (0.890)	−0.026*** (0.030)	−15.211*** (6.690)	0.066
Hokuriku	0.446*** (0.036)	−68.197*** (6.889)	2.411*** (0.228)	481.420*** (51.951)	0.280
Kyushu	0.531*** (0.035)	78.699*** (8.552)	−2.277*** (0.264)	−675.556*** (69.237)	0.293
Shikoku	0.226*** (0.039)	16.276*** (6.476)	−0.376* (0.215)	−158.613*** (48.686)	0.312
Tohoku	0.585*** (0.049)	−27.757*** (6.474)	0.992*** (0.201)	191.624*** (51.998)	0.252
Tokyo	0.698*** (0.052)	−17.800*** (4.318)	0.625*** (0.125)	122.857*** (37.384)	0.321

Standard errors are in parentheses, $^{*}p < 0.1$, $^{***}p < 0.01$.

Comparative analysis under different models. This part compared the results of the carbon pass-through rate at different models. Based on the regression returns of the mixed model in Table 4, Japan's average carbon cost pass-through rate is 0.596, statistically significant at a 5% level. It means that if carbon cost increased by 1 unit, about 60% of one unit would be switched to electricity price rises. However, electricity demand and constant terms are not statistically significant, and they may arise from the fixed effects among regions, and the variations in each region also harm the robustness of the average value of CPTR.

In order to testify to the robustness of CPTR under the nonlinear relationship between fuel spread and electricity demand besides the quadratic OLS regression method, this paper also applies the generalized additive model (GAM). GAM can flexibly capture the nonlinear relationship on many shapes and assumes a Gaussian distribution of the error term with zero mean and variance. The smooth function for independent variable demand is $f(D_i)$. The regression returns are shown in Table 5. Edf stands for effective degrees of freedom for the smooth functions. This value represents the complexity of the smooth—all edf in returns higher the 2, which is the quadratic curve. The CPTRs in the GAM are statistically significant at 1%, and values are also approximately the same as the polynomial OLS regression results. Thus, the estimations under polynomial regression model are robust among regions. Kansai, Chubu, and Chugoku still have a higher carbon pass-through rate than the rest regions. Hokuriku, Kyushu, Shikoku, Tohoku, and Tokyo have a lower carbon pass-through rate. However, the CPTR in Kyushu is 0.077 in GAM, much smaller than the polynomial regression, which is 0.531. The possible explanation could be that the fuel spread in Kyushu has

Table 4 Linear regression mixed model $P^e - P^f = \alpha_0 + \alpha_1 C + \alpha_2 D + \varepsilon$.

Price	Coef.	St. err.	t-value	p-value	[95% conf interval]		Sig
Demand	0.173	0.127	1.36	0.210	−0.120	0.467	
Carbon_cost	0.596	0.196	3.03	0.016	0.143	1.049	**
Constant	−1.007	1.765	−0.57	0.584	−5.076	3.063	
Mean-dependent var		0.705	SD-dependent var			1.352	
R-squared		0.026	Number of obs.			258,290	
F-test		20.099	Prob > *F*			0.001	
Akaike crit. (*AIC*)		882,109.374	Bayesian crit. (*BIC*)			882,140.759	

**$p < 0.05$.

Table 5 Regression results from the generative additive model $P_i^e - P_i^f = \alpha_0 + \alpha_1 C_i + f(D_i) + \varepsilon$.

Regions	α_1	α_0	$f(D_i)$		R^2
			edf	p-value	
Kansai	1.962*** (0.038)	4.577*** (0.078)	8.046	<2e-16***	0.368
Chubu	2.153*** (0.053)	5.226*** (0.114)	7.743	<2e-16***	0.319
Chugoku	3.004*** (0.084)	6.123*** (0.155)	7.254	<2e-16***	0.292
Hokkaido	−0.051*** (0.003)	0.656*** (0.005)	8.717	<2e-16***	0.098
Hokuriku	0.458*** (0.036)	1.460*** (0.070)	6.222	<2e-16***	0.281
Kyushu	0.077*** (0.014)	0.708** (0.030)	6.798	<2e-16***	0.788
Shikoku	0.225*** (0.039)	1.005*** (0.076)	8.538	<2e-16***	0.314
Tohoku	0.360*** (0.208)	1.264*** (0.041)	7.527	<2e-16***	0.093
Tokyo	0.728*** (0.052)	2.241*** (0.113)	8.397	<2e-16***	0.323

**$p < 0.05$, ***$p < 0.01$.

Table 6 Comparative analysis for the key indicator (carbon pass-through rate) among different models.

Regions	Polynomial OLS		Linear OLS		GAM	
	Estimated CPTR	R^2	Estimated CPTR	R^2	Estimated CPTR	R^2
Kansai	1.937***	0.365	1.947***	0.365	1.962***	0.368
Chubu	2.157***	0.318	2.061***	0.315	2.153***	0.319
Chugoku	2.995***	0.291	2.895***	0.286	3.004***	0.292
Hokkaido	−0.055***	0.066	−0.052***	0.065	−0.051***	0.098
Hokuriku	0.446***	0.28	0.411***	0.277	0.458***	0.281
Kyushu	0.531***	0.293	0.576***	0.291	0.077***	0.788
Shikoku	0.226***	0.312	0.230***	0.312	0.225***	0.314
Tohoku	0.585***	0.252	0.611***	0.252	0.360***	0.093
Tokyo	0.698***	0.321	0.667***	0.320	0.728***	0.323

Polynomial OLS model: $P_i^e - P_i^f = \alpha_0 + \alpha_1 C_i + \beta_1 D_i + \beta_2 D_i^2 + \varepsilon$.
Linear OLS model: $P_i^e - P_i^f = \alpha_0 + \alpha_1 C_i + \alpha_2 D_i + \varepsilon$.
GAM: $P_i^e - P_i^f = \alpha_0 + \alpha_1 C_i + f(D_i) + \varepsilon$.
***$p < 0.01$.

many negative values and that the higher degree of the nonlinear relationship is a finer pattern (Table 6)[3]

Conclusions

Before the electricity price fully reveals the increment of carbon cost, the cost from emissions should be fully internalized into the cost of production. Ideally, carbon pricing should entirely capture the social cost caused by CO_2 emissions, a low level of carbon tax rate cannot fully reveal the social costs. For Japan, the current carbon tax rate is well below European countries. Therefore, carbon pricing cannot fully reflect social costs under electricity generation. Secondly, the carbon cost pass-through rate varies among regions. Each region introduced its programs at a federal level to achieve the goal of carbon neutrality. For example, Tokyo introduced a CO_2 emissions-related program at the federal level in 2010, TMG ETS. Instead of voluntary participants like Japan Voluntary Emissions Trading Scheme nationwide, this scheme in Tokyo is mandatory for firms and end-users. The program is promising in constantly reducing emissions compared with JVET. However, firms with identical carbon intensity not covered by the same level of carbon cost would result in carbon leakage from less regulated areas to high-carbon cost areas. Firms in less regulated areas may be more competitive on the product's price than firms in high-carbon cost areas. Consequently, it drags down the reduction of emissions. Therefore, Japan should continue nationwide carbon tax reformation by the rising carbon tax rate to assist with carbon neutrality.

The other prerequisite for a full carbon cost pass-through is electricity liberalization. Market-clearing price equals marginal cost, including carbon cost in a perfectly competitive market, and suppliers cannot manipulate price settings. In Japan, only suppliers and customers who meet requirements on trading abilities are allowed to enter the wholesale market. Incumbent utilities actively participate in electricity sales and purchases at JEPX and account for about 75% of trading volume (Ofuji, 2016). Consequently, electricity spot market price is not reliable for analyzing the correlation between price and demand, supply and cost. Although deregulation in the retail market made progress in 2016 as entry limitations for suppliers and consumers were removed, electricity liberalization in Japan is still in process. Trading volume in the retail market expanded, stimulating the wholesale market's ban-lifting process. However, many issues remain unsolved for a stable electricity supply to meet demand from end-users. Adjusting the carbon tax rate and continuing electricity liberalization are the two crucial points for Japan to reduce CO_2 emissions in the electricity sector and thus achieve carbon neutrality by 2050.

Data availability

The datasets generated during and/or analyzed during the current study are available from the corresponding author on reasonable request.

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Notes

1 The three major factors for a power plant are the carbon pricing mechanism, energy efficiency, and renewable efficiency. The model in this paper includes the carbon tax for the carbon pricing mechanism and the heat rate for energy efficiency. The reason that does not include renewable energy in the model is that for most regions in Japan, electricity generation still heavily relies on traditional inputs, such as natural gas and coal, which account for about 65% of total inputs. However, future research should consider renewable efficiency when renewable resources account for a high proportion of power generation processes. Besides, the electricity price here is assumed at the equilibrium point when demand equals the supply.

2 According to the Energy Information Association, the spread or dark spark is calculated using the following formulation: Spark (dark) spread = power price – [gas (coal) price * heat rate].

3 The comparative results for carbon pass-through rate under different models are listed in Table 6.

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The author declares no competing interests.

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Additional information

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